

# Automated Safety Stock & Inventory Optimization System

*A Dynamic ABC XYZ Reorder Point & Working Capital Model for a Multi Market Distribution Network*

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## Executive Summary

- Built a dynamic, SKU level safety stock and reorder point system on the DataCo Supply Chain dataset (180,519 order line records, 118 active SKUs, Jan 2015 – Jan 2018) using SQL (MySQL) and Power BI/DAX only.
- ABC (revenue) × XYZ (demand volatility) segmentation found zero stable demand SKUs and 109 of 118 (92%) classified erratic (Z class,  $CV > 0.5$ ) directly motivating a dynamic, not static, safety stock policy.
- A combined variance safety stock model drives every KPI as a live function of the Service Level slicer: moving 90% → 99% shifts safety stock working capital \$316.8K → \$576.7K and reorder breaches 11 → 33 SKUs (Table I) quantifying, not just asserting, the cost of tighter fill rate targets.
- Identified and corrected three data integrity defects during development a lead time unit mismatch (~16× reorder point inflation), a revenue period misalignment (~3.1× working capital overstatement), and a source data completeness gap each found by validating model output against source data rather than assuming correctness.

**Index Terms** *Safety Stock Optimization, ABC XYZ Classification, Reorder Point, Economic Order Quantity, Power BI, DAX, SQL, Supply Chain Analytics, Inventory Management*

## 1. Executive Summary & Business Problem

### Business Context

The analysis covers a multi market e-commerce distribution operation trading across five global markets LATAM, Europe, Pacific Asia, USCA (North America), and Africa drawn from the DataCo Supply Chain dataset (180,519 order line transactions, January 2015 – January 2018, 118 active SKUs). The business operates a single echelon replenishment model with no automated reorder logic; stock decisions are made on static, catalog wide reorder rules rather than SKU level demand and lead time behaviour.

### The Bleed / Bottleneck

XYZ segmentation of the full 118 SKU catalog found zero SKUs with stable, low variability demand (X class): 9 SKUs fall into moderate variability (Y class) and 109 92% of the catalog are erratic (Z class, coefficient of variation  $> 0.5$ ). A static, one size fits all safety stock policy is structurally mismatched to this demand profile. At a 90% service level and 2.4× simulated stock coverage, 11 SKUs (9.32%) have already breached their reorder point; tightening to 99% recalculating every SKU's own statistical trigger, not the on hand assumption raises this to 33 SKUs (27.97%), since a higher service level demands a higher buffer even where current stock hasn't moved.

### Objective

Replace the static reorder policy with a dynamic, service level driven safety stock and reorder point model that: (1) right sizes safety stock investment per SKU using combined demand and lead time variability, (2) flags at risk SKUs before stockout with a clear expedite / monitor / no action signal, and (3) lets operations leadership trade off working capital cost against fill rate risk in real time via an interactive service level slicer.

## 2. Data Architecture & Engineering Constraints

### Data Sources & Schema

Single source table `orders_raw` (53 columns) loaded from the DataCo Supply Chain CSV [1] into MySQL 8.0. Grain is one row per order line item, carrying order/shipping dates, customer and product dimensions, quantities, pricing, profit, delivery status, and shipping mode. From this, a derivation chain builds: `orders_clean` (fraud/cancel filtered) → `calendar_months` (37 month spine) → `monthly_demand` (118 SKUs × 37 months, zero filled) → `demand_stats / leadtime_stats` (per SKU mean & standard deviation) → `abc_class` (revenue Pareto) → `xyz_class` (demand coefficient of variation) → `sku_master`, the 118 row analytical base table exported to Power BI.

### Assumptions & Edge Cases

- No live inventory feed exists in the source data, so “current stock” is modelled as a simulated inventory position (average monthly demand × an adjustable coverage multiplier) due to the absence of a warehouse/ERP feed not an actual on hand balance flagged explicitly on every dashboard view.
- `CANCELED` and `SUSPECTED_FRAUD` orders (7,754 rows) were excluded before any aggregation: these map to shipments that never actually left the warehouse and would otherwise inflate demand volume and fabricate lead time durations.
- A genuine data completeness cliff exists from October 2017 onward (active SKU count falls from 50+ to under 20, confirmed against raw order volume, not a pipeline defect) the trend and forecast basis is truncated to complete months only.
- Lead time is recorded in days while demand is recorded in units/month; every safety stock and ROP formula performs an explicit  $\div 30$  unit conversion before combining the two variance terms.

### Data Cleaning & Pipeline

Ingestion and transformation were performed entirely in SQL: a recursive CTE generates the zero filled month spine (required so months with no sales register as explicit zero demand rows rather than silently understating volatility), window functions compute cumulative revenue share for ABC classification, and STDDEV\_SAMP / AVG drive the XYZ coefficient of variation. The resulting sku\_master and monthly\_demand tables were imported live into Power BI, where DAX measures layer on the what if and risk scoring logic (Section 3).

### 3. Technical Execution & Methodology

### Core Logic / Algorithmic Approach

Two complementary segmentation and risk scoring layers were used:

- ABC (revenue Pareto): SKUs ranked by cumulative revenue share A = top 80%, B = next 15%, C = bottom 5% (result: A=7, B=16, C=95 SKUs).
- XYZ (demand volatility): coefficient of variation of monthly demand  $X < 0.2$ ,  $0.2 \leq Y \leq 0.5$ ,  $Z > 0.5$  (result: X=0, Y=9, Z=109 SKUs).
- Combined variance safety stock [2]:  $SS = Z \times \sqrt{(L \cdot \sigma_d^2 + \bar{D}^2 \cdot \sigma_L^2)}$ , incorporating both demand and lead time variability.

**Caveat flagged at formula introduction:** this formula assumes normally distributed demand. With 109 of 118 SKUs (92%) classified Z class ( $CV > 0.5$ ), demand for most of the catalog is intermittent rather than normal, and the Gaussian assumption understates true tail risk safety stock for these items. It also assumes  $Cov(Demand, Lead\ Time) = 0$  in a multi market operation, demand surges and carrier delays can plausibly correlate (e.g. peak season demand coinciding with the same congestion that slows shipping), which would make the variance term an underestimate. Both limitations are addressed fully in Section 6.

- Reorder Point:  $ROP = (\bar{D} \times L) + \text{Safety Stock}$ , with the Z factor driven live by a 90/95/99% service level slicer the single input that drives every range reported in Section 5.
- EOQ (Wilson formula) [3]:  $\sqrt{(2 \times \text{Annual Demand} \times \text{Order Cost} \div \text{Holding Cost per Unit})}$ , unit price reconstructed from  $\text{total\_revenue} \div (\text{avg\_demand} \times 37)$ .

### Code / Model Artifacts

SQL ABC/XYZ classification (MySQL 8.0, window functions):

```
with revenue_by_sku as (
  select product_card_id, sum(sales) as total_revenue
  from orders_clean
  group by product_card_id
),
ranked as (
  select product_card_id, total_revenue, sum(total_revenue) over (order by total_revenue desc) / sum(total_revenue)
  over () as cum_revenue_pct
  from revenue_by_sku
)

select product_card_id, total_revenue,
case
  when cum_revenue_pct <= 0.80 then 'a'
  when cum_revenue_pct <= 0.95 then 'b'
  else 'c'
end as abc_class
from ranked;
```

DAX combined variance Safety Stock, and the PO Priority Tier risk classifier (Power BI):

```
safety stock (units) =
var z = [service level z]
return
sumx(sku_master, var lead_time_months = sku_master[avg_leadtime] / 30
      var std_dev_lead_time_months = sku_master[stddev_leadtime] / 30
      return
      z * sqrt((lead_time_months * sku_master[stddev_demand] ^ 2) + (sku_master[avg_demand] ^ 2 * std_dev_lead_time_months
      ^ 2)))

po priority tier =
var z = [service level z]
return
sumx(sku_master, var lead_time_months = sku_master[avg_leadtime] / 30
      var std_dev_lead_time_months = sku_master[stddev_leadtime] / 30
      var safety_stock =
```

```

z * sqrt((lead_time_months * sku_master[stddev_demand] ^ 2) + (sku_master[avg_demand] ^ 2 * std_dev_lead_time_months ^ 2))
var rop = (sku_master[avg_demand] * lead_time_months) + safety_stock
var on_hand = sku_master[avg_demand] * [coverage]
return
switch(
true(), on_hand <= safety_stock, 1,
on_hand <= rop, 2,
on_hand <= rop * 1.15, 3, 4) )

```

PO Priority Tier is a scalar measure, not a column: inside FILTER or SUMX (as used in Reorder Breach Count and the alert table), DAX context transition evaluates it correctly per SKU, so the current dashboard's totals are not summing tier codes. It is, however, unsafe to drag directly into a Card visual with no SKU level filter doing so would sum categorical tier codes and ROP quantities across the whole catalog, a real but easy mistake for a future collaborator to make. Catalog Worst Tier (MINX, above) is included as the aggregate safe alternative for any future top level KPI.

## 4. Dashboards, Simulators & Visual Deliverables

### Operational Dashboard

The primary Power BI view (Fig. 1) surfaces catalog wide risk at a glance: total safety stock working capital, reorder breach count, revenue at risk, average lead time, and a % SKUs at risk gauge, alongside an ABC × XYZ revenue at risk matrix and a ranked SKU alert table (Stock to ROP ratio with a red/amber/green PO Action column: Emergency Expedite, Standard Reorder, Monitor, No Action).



Figure 1. Inventory Risk & Safety Stock Optimization Dashboard, captured at 95% service level / 20% holding cost / \$170 order cost / 2.4× coverage multiplier one point on the range quantified in Table I.

### User Interface / Interaction

An operations manager works top down: adjust the Service Level tile (90/95/99%), Holding Cost %, Order Cost, and Coverage Multiplier sliders to model a scenario, then drill through any alert table row into a per SKU Action Center (Fig. 2) showing simulated stock, safety stock, EOQ, days of supply, annual holding/ordering cost, revenue at risk, and a 37 month demand trend.



Figure 2. SKU Action Center drill through per SKU safety stock, EOQ, cost, and demand trend detail, slicer state synchronized to the main dashboard.

## 5. Quantified Business Impact & ROI

Every KPI on the dashboard is a function of the Service Level slicer; reporting a single snapshot would misrepresent a model built specifically for scenario comparison. Table I holds the Coverage Multiplier fixed at the Figure 1 value (2.4×) and varies only Service Level:

| Service Level | SS Working Capital | Reorder Breaches | Revenue at Risk | % SKUs at Risk |
|---------------|--------------------|------------------|-----------------|----------------|
| 90%           | \$316,799          | 11               | \$366,452       | 9.32%          |
| 95%           | \$407,136          | 29               | \$2,403,377     | 24.58%         |
| 99%           | \$576,673          | 33               | \$2,574,486     | 27.97%         |

TABLE I. SERVICE LEVEL SENSITIVITY OF SAFETY STOCK CAPITAL AND REORDER RISK (COVERAGE MULTIPLIER FIXED AT 2.4×)

### Primary Metric

Moving from 90% to 99% service level nearly triples reorder breaches (11 → 33 SKUs) and increases safety stock working capital by 82% (\$316.8K → \$576.7K). Revenue at risk jumps sharply between 90% and 95% (\$366K → \$2.40M) because several mid tier SKUs sit just above the 90% threshold but below the 95% one a genuine cliff in the data, not a rounding artifact.

### Financial Metric

Safety stock working capital is not a fixed cost of the model it is a direct, adjustable output of the chosen service level, visible and quantified rather than implicit in a static reorder rule. Table I lets management see the exact capital cost of each fill rate point before committing to it. The ABC × XYZ risk matrix further shows revenue at risk concentrated in B class and C class, Z class SKUs, meaning the highest revenue A class items are comparatively well protected across the full service level range tested.

## 6. Limitations & Scalability Roadmap

### Edge Case Vulnerabilities

- The combined variance safety stock formula assumes normally distributed demand. With 109 of 118 SKUs (92%) in the erratic Z class ( $CV > 0.5$ ), demand for most of the catalog is intermittent/lumpy rather than normal, and the normal distribution assumption understates true safety stock at the tails for these items. For production use on Z class SKUs, Poisson or negative binomial demand models, or Croston's method [4] for intermittent demand forecasting, are the standard alternative and were scoped out of this build.
- The formula also assumes demand and lead time are statistically independent ( $Cov(D, L) = 0$ ). In a multi market fulfillment operation this is a simplification: demand surges (e.g. peak season) can plausibly coincide with carrier congestion and longer lead times, which would make the true combined variance and therefore the required safety stock larger than this model estimates.
- Simulated Stock Level is a static Coverage Multiplier × average demand proxy, not a real inventory position explicitly a simulated inventory position adopted due to the absence of a warehouse/ERP feed in the source data.

A uniform multiplier also ignores SKU specific replenishment cycles and can create artificial breach clustering.

- The Oct 2017 – Jan 2018 data completeness cliff means the most recent ~4 months are excluded from trend measures; demand averages are therefore weighted toward 2015–2017 behaviour.
- Single echelon model: lead time and demand are analysed at the SKU level only, with no distribution center to store or supplier tier decomposition.

### Next Iterations

- Replace the average/standard deviation demand model with Croston's method or a negative binomial demand model specifically for the 109 Z class SKUs.
- Model demand/lead time correlation explicitly (e.g. a copula or conditional lead time distribution keyed to demand percentile) rather than assuming independence.
- Integrate a live inventory feed (WMS/ERP API) to replace the simulated stock proxy with actual on hand balances.
- Extend from single echelon to multi echelon inventory optimization across distribution centres, and automate the Emergency Expedite / Standard Reorder flags into a Power Automate or ERP integrated PO trigger.

## 7. Conclusion

This project demonstrates an end to end path from a raw, uncleaned transactional export to a live, decision ready inventory risk model, built entirely in SQL and Power BI/DAX. The most consequential findings were not the final dashboard numbers but the debugging process that produced them: a lead time unit mismatch that had inflated reorder points roughly sixteen fold, and a revenue period misalignment that had overstated safety stock working capital by approximately 3.1×, were both identified by systematically validating model output against source data. Because every KPI is service level driven, the model's real deliverable is not a single number but a quantified range Table I that converts an undifferentiated 118 SKU catalog into a ranked, adjustable action list, and provides a template that scales conceptually toward the extensions outlined in Section 6.

## 8. References

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